November 13, 2026

To the Editor-in-Chief and Editorial Board of *Renewable Energy*

Dear Editor,

We are pleased to submit our original research article entitled *"Scale-Bridging Techno-Economic, Microclimatic, and Open-Quantum Assessment of Semi-Transparent Organic Agrivoltaics for Sub-Saharan Perennial Crops"* for consideration in *Renewable Energy*.

Wavelength-selective semi-transparent organic photovoltaics (ST-OPVs) are widely proposed for tropical agrivoltaics on the premise that transmitted spectra enhance crop productivity relative to opaque crystalline silicon (c-Si) while modulating photosynthetic quantum coherence. Existing evaluations rarely link calibrated transfer-matrix optics, hourly tropical microclimate and stomatal–soil-water physiology, cooperative processing dispatch, replacement-aware life-cycle accounting, and global uncertainty propagation within one self-consistent counterfactual framework. Here, we assess three flagship perennial production chains across western Cameroon's wet–dry highland belt—Arabica coffee (Foumbot, 1,044 m), Protected Geographical Indication (PGI) Penja white pepper (Njombe-Penja, 273 m), and highland tea (Djuttitsa, 1,400 m)—over a full 8,760 h (2023 ERA5) meteorological cycle.

A four-lever Shapley decomposition of the modelled yield shifts attributes a pronounced seasonal regime inversion: dry-season leaf cooling (-1.5 to $-3.5\,^{\circ}\mathrm{C}$) boosts dry-season assimilation across all species (+8.8% to +30.2% rainfed; +49.7% irrigated), whereas monsoon cloudiness drives net annual yields negative in coffee (-7.08%) and tea (-3.35%) under 40.6% shade, leaving Penja white pepper as the sole crop sustaining net rainfed gains (+7.43% to +8.16%, land equivalent ratio 1.48–1.55). At matched $\sim 40\%$ shade, ST-OPV and elevated c-Si crop yields differ by at most 1.05 percentage points, so within the model shade fraction and seasonal timing, not spectral selectivity, govern agronomy. Multi-Hamiltonian continuous-wave steady-state bounds (cross-checked against MesoHOPS 1.8.0 pulsed propagation and 4-site exact HEOM) constrain excitonic efficiency shifts below 3.37×10^{-5} (2.10×10^{-3} HEOM envelope), resulting in negligible Sobol sensitivity ($S_{Ti} \leq 2.60 \times 10^{-7}$).

Across $N = 10,000$ Saltelli quasi-Monte Carlo realizations, dual variance decomposition shows that grid export tariffs and cost of capital govern standalone ST-OPV bankability, whereas hardware costs ($C_{\mathrm{mod}}^{\mathrm{OPV}}, C_{\mathrm{BOS}}^{\mathrm{Si}}$) and encapsulation lifetime (T_{80}) govern competition against elevated c-Si, which achieves higher 20-year NPV in 77.0%–89.0% of draws (75.3%–87.7% at zero carbon). Life-cycle carbon parity requires $T_{80,\mathrm{LCA}}^{*} \geq 7.40$–$8.12$ yr (slashing intensity to 7.8–12.5 $\mathrm{gCO_2e/kWh}$ at 25 yr), whereas economic NPV parity emerges only in Penja pepper at $\$15/\mathrm{m}^2$ with $T_{80,\mathrm{NPV}}^{*} \geq 16.33$ yr; overnight tea withering storage lifts solar fraction to 99.9% and sustains positive off-grid NPV ($+\$319,794$/ha).

The manuscript runs to 21 pages—4,567 words of main text (5,886 including captions) with 50 references—and is accompanied by a 9-page Supporting Material, in line with the journal's length and reference guidance.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors approve this submission and declare no competing financial interests.

Sincerely,

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